High dietary antioxidant intakes are associated with decreased chromosome translocation frequency in airline pilots

Background: Dietary antioxidants may protect against DNA damage induced by endogenous and exogenous sources, including ionizing radiation (IR), but data from IR-exposed human populations are limited. Objective: The objective was to examine the association between the frequency of chromosome translocations, as a biomarker of cumulative DNA damage, and intakes of vitamins C and E and carotenoids in 82 male airline pilots. Design: Dietary intakes were estimated by using a self-administered semiquantitative food-frequency questionnaire. Translocations were scored by using fluorescence in situ hybridization with whole chromosome paints. Negative binomial regression was used to estimate rate ratios and 95% CIs, adjusted for potential confounders. Results: Significant and inverse associations were observed between translocation frequency and intakes of vitamin C, β-carotene, β-cryptoxanthin, and lutein-zeaxanthin from food ($P < 0.05$). Translocation frequency was not associated with the intake of vitamin E, α-carotene, or lycopene from food; total vitamin C or E from food and supplements; or vitamin C or E or multivitamin supplements. The adjusted rate ratios (95% CI) for ≥median compared with <median servings per week of high–vitamin C fruit and vegetables, citrus fruit, and green leafy vegetables were 0.61 (0.43, 0.86), 0.64 (0.46, 0.89), and 0.59 (0.43, 0.81), respectively. The strongest inverse association was observed for ≥median compared with <median combined intakes of vitamins C and E, β-carotene, β-cryptoxanthin, and lutein-zeaxanthin from food: 0.27 (0.14, 0.55). Conclusion: High combined intakes of vitamins C and E, β-carotene, β-cryptoxanthin, and lutein-zeaxanthin from food, or a diet high in their food sources, may protect against cumulative DNA damage in IR-exposed persons.